Edgar J. Andrade-Lotero, Ph.D.

edgar.andrade@urosario.edu.co | LinkedIn | GitHub

EXPERIENCE

Postdoctoral Scholar 2025 – Present
University of California, Davis, USA *Department of Communication*
Computational Communication Research Lab (Seth Frey). Behavioral experiments and computational models of coordination and resource allocation in groups.

Universidad del Rosario, Bogotá, Colombia 2006 – Present
Associate Professor, School of Sciences and Engineering 2019 – Present
Associate Professor, Faculty of Natural Sciences and Mathematics 2010 – 2019
Auxiliary Professor, School of Social Sciences July 2006 – 2010

Lecturer February 2006 – June 2006
Universidad Sergio Arboleda, Bogotá, Colombia *School of Mathematics*

Lecturer February 2004 – June 2004
Universidad Sergio Arboleda, Bogotá, Colombia *School of Mathematics*

Lecturer 2003 – 2004
Universidad Nacional de Colombia, Bogotá, Colombia *Department of Mathematics*

Research collaborations: Percepts and Concepts Laboratory, Robert Goldstone (Indiana University); Computational Cognition Lab, Nori Jacoby (Cornell University).

EDUCATION

University of Amsterdam
Ph.D. in Logic. Thesis: Models of Language: Towards a practice-based account of information in natural language. 2012

University of Amsterdam
M.Sc. in Logic. Thesis: Meaning and Form in Event Calculus. 2006

Universidad Nacional de Colombia
B.A. in Mathematics. 2002

SELECTED PUBLICATIONS

- R. L. Goldstone, **E. J. Andrade-Lotero**, R. Hawkins, and M. Roberts, "The emergence of specialized roles within groups," Topics in Cognitive Science, vol. 16, pp. 257–281, 2023.
- **E. J. Andrade-Lotero**, J. M. Ortiz-Duque, J. A. Velasco-García, and R. L. Goldstone, "The division of linguistic labour for offloading conceptual understanding," Philosophical Transactions of the Royal Society B, vol. 378, no. 20210360, 2023.
- **E. J. Andrade-Lotero** and R. L. Goldstone, "Self-organized division of cognitive labor," PLOS ONE, vol. 16, no. 7, 2021.
- **E. J. Andrade-Lotero**, C. Álvarez, and G. Obando, "A paradox in Brian Arthur's solution to his El Farol Bar problem," Advances in Complex Systems, vol. 27, no. 8, p. 2550014, 2025.
- **E. J. Andrade-Lotero** and R. L. Goldstone, "Self-organized division of cognitive labor," Proceedings of the 41st Annual Conference of the Cognitive Science Society, 2019, pp. 91–97.

PUBLICATIONS: RESEARCH ARTICLES

- **E. J. Andrade-Lotero**, R. L. Goldstone, and S. Frey, "Iterative cognitive schema matching explains open-ended coordination," manuscript submitted for publication, 2026.
- D. Yela, G. Obando, A. Pantoja, **E. J. Andrade-Lotero**, and J. Barreiro-Gomez, "Finding the delay that maximizes the convergence rate of a class of LTI systems," Journal of the Franklin Institute, vol. 363, no. 15, p. 109012, 2026.
- **E. J. Andrade-Lotero** and J. Ortiz-Duque, "A mis-conception of logic in its diffusion across non-mathematical environments," Linguistic and Philosophical Investigations, vol. 18, pp. 77–96, 2019.
- C. Álvarez, **E. J. Andrade-Lotero**, and V. Gauthier, "Modelos epidemiológicos en redes: Una presentación introductoria" (Epidemiological models on networks: an introductory presentation), Boletín de Matemáticas, vol. 22, no. 1, pp. 21–37, 2015.
- **E. J. Andrade-Lotero**, "Una gramática lógica para explicar algunos aspectos de la semántica del español" (A logical grammar for some aspects of Spanish semantics), Boletín de Matemáticas, vol. 20, no. 1, pp. 21–50, 2013.
- **E. J. Andrade-Lotero** and C. Dutilh, "A contentious trinity: Levels of entailment in Brandom's pragmatic inferentialism," Philosophia, vol. 40, no. 1, pp. 41–53, 2012.

- **E. J. Andrade-Lotero** and C. Dutilh, "Validity, the squeezing argument and alternative semantic systems: The case of Aristotelian syllogistic," Journal of Philosophical Logic, vol. 41, no. 2, pp. 387–418, 2012.
- **E. J. Andrade-Lotero** and E. Becerra, "Establishing connections between Aristotle's natural deduction and first-order logic," History and Philosophy of Logic, vol. 29, no. 4, pp. 309–325, 2008.
- **E. J. Andrade-Lotero** and E. Becerra, "Corcoran's Aristotelian syllogistic as a subsystem of first-order logic," Revista Colombiana de Matemáticas, vol. 41, no. 1, pp. 67–80, 2007.

PUBLICATIONS: BOOKS AND TRANSLATIONS

- L. Gamut, Lógica, lenguaje y significado: lógica intensional y gramática lógica (Logic, Language, and Meaning: Intensional Logic and Logical Grammar), **E. Andrade-Lotero** and C. M. (translators), Eds. Bogotá: Editorial Universidad del Rosario, 2010.
- **E. J. Andrade-Lotero**, P. Cubides, C. Márquez, E. J. Vargas, and D. Cancino, Lógica y Pensamiento Formal (Logic and Formal Thought). Editorial Universidad del Rosario, 2008.

PUBLICATIONS: CONFERENCE PROCEEDINGS (EDITED)

- M. S. Dora Achouroti, **Edgar J. Andrade-Lotero**, Eds., Proceedings of the Amsterdam Graduate Philosophy Conference: Meaning and Truth, X-2010-01, University of Amsterdam, ILLC Publications, 2010.
- M. S. Dora Achouroti, **Edgar J. Andrade-Lotero**, Eds., Proceedings of the Graduate Philosophy Conference on Normativity, X-2009-02, University of Amsterdam, ILLC Publications, 2009.

PUBLICATIONS: CONFERENCE PAPERS

- **E. J. Andrade-Lotero**, O. Tchernichovski, N. Jacoby, D. Conley, and S. Frey, "Cultural evolution of power dynamics between workers and managers in a collective foraging task," poster, Proceedings of the 48th Annual Meeting of the Cognitive Science Society, 2026.
- **E. J. Andrade-Lotero**, O. Tchernichovski, N. Jacoby, D. Conley, and S. Frey, "Reinforcement learning with people helps overcome conflicts of interest in social dynamics," poster, International Conference on Computational Social Science (IC2S2), 2026.
- J. Piza-Londoño and **E. J. Andrade-Lotero**, "Leveraging semantic parsing using text embeddings and reinforcement learning," 2024 IEEE Latin American Conference on Computational Intelligence (LA-CCI), pp. 1–6, 2024.
- **E. J. Andrade-Lotero** and R. L. Goldstone, "An empirical study of the El Farol Bar problem," Collective Intelligence, 2024.
- **E. J. Andrade-Lotero** and R. L. Goldstone, "Self-organized division of cognitive labor," Collective Intelligence, 2019.
- **E. J. Andrade-Lotero** and J. Ortiz, "The proper approach to the division of linguistic labor," Sixth International Conference on Philosophy of Language and Linguistics, 2019.
- **E. J. Andrade-Lotero**, "Translations and structures: Some reflexions about teaching the semantics of first-order logic," Philogica IV, 2016.
- **E. J. Andrade-Lotero**, "Explicating compositionality," Philogica III, 2014.
- **E. J. Andrade-Lotero**, "On truth conditions that depend on performatives," Third International Conference on Philosophy of Language and Linguistics, 2013.
- **E. J. Andrade-Lotero**, "El principio de composicionalidad y la competencia lingüística" (The principle of compositionality and linguistic competence), IV Congreso Colombiano de Filosofía, 2012.
- **E. J. Andrade-Lotero**, "Compositionality from a 'use-theoretic' perspective," 32nd International Wittgenstein Symposium, 2009.
- **E. J. Andrade-Lotero** and C. Dutilh, "A reconstruction of the hierarchical relations between incompatibility entailment, committive and permissive consequence," Proceedings of the Workshop on Bob Brandom's Recent Philosophy of Language: Towards an Analytic Pragmatism, 2009, pp. 90–98.

STUDENTS SUPERVISED

Titles of theses written in Spanish are given in English.

- J. V. Tamayo-Ramírez, A general framework for designing and evaluating intelligent tutoring systems: using large language models rigorously, M.Sc. in Engineering, Science and Technology, 2026.
- J. P. Rivera Burgos, Making the invisible visible: a natural language processing tool to systematize news about homicides of LGBTIQ+ people, M.Sc. in Applied Mathematics and Computer Science, 2025.
- C. A. Gómez Vargas, Integrating reinforcement learning into semantic parsers for logical deduction from natural language, M.Sc. in Applied Mathematics and Computer Science, 2025.

- S. Carrerá Martínez, Points and lines: applying graph theory to the behavior of Colombian political parties in Congress, M.Sc. in Applied Mathematics and Computer Science, 2025.
- E. J. López-López and S. Gaitán-Rubio, Modeling prices with the minority game, M.Sc. in Applied Mathematics and Computer Science, 2024.
- E. Muñoz, Active management of crypto-asset portfolios using reinforcement learning, M.Sc. in Applied Mathematics and Computer Science, 2024.
- J. Piza-Londoño, Reinforcement learning of an optimal semantic parser in Discourse Representation Theory, M.Sc. in Applied Mathematics and Computer Science, 2024. Published as Piza-Londoño and Andrade-Lotero, LA-CCI 2024.
- J. V. Tamayo-Ramírez, G. A. N. Cárcamo, and R. D. Rico-González, Automating IQAP qualification processes with an AI chatbot for admission to international universities, M.Sc. in Applied Mathematics and Computer Science, 2024.
- J. A. Velasco, Toward a cartography of meaning: the roles of sense in the *Tractatus Logico-Philosophicus*, B.A. in Philosophy, 2019.
- M. P. Gómez, The incompleteness of the foundations of mathematics, B.A. in Philosophy, 2018.
- M. G. Algarra, Incommensurability and interpretation: a proposal for comparing scientific theories, B.A. in Philosophy, 2007.

COURSES TAUGHT

- **Computer science**, Universidad del Rosario. Graduate, 2021–2024: Reinforcement Learning; Artificial Intelligence: Problem Representation and Solving; Machine Learning Applications. Undergraduate, 2017–2024: Artificial Intelligence: Problem Representation and Solving; Logic for Computer Science; Theory of Computation; Numerical Analysis and Scientific Computing. Undergraduate, 2007: Formal Methods; Processes and Algorithms.
- **Logic and philosophy.** Universidad del Rosario, undergraduate, 2007–2016: Philosophy of Language; Logic; Logical-Philosophical Models of Natural Language; Pragmatics. Mathematical Logic, Universidad Sergio Arboleda, 2003. Logic for Psychologists, Universidad Nacional de Colombia, 2002.
- **Mathematics**, Universidad del Rosario, except as noted. Undergraduate, 2017–2024: Linear Algebra; Logic, Number Theory and Sets; Logic and Discrete Mathematics. Undergraduate, 2007–2016: Integral Calculus; Probability; Inferential Statistics.

CERTIFICATIONS

- Deep Learning, Coursera
- Reinforcement Learning Specialization, Coursera
- Machine Learning Engineering for Production, Coursera
- Generative AI with Large Language Models, Coursera

SKILLS

Programming Languages: Python, PyTorch, R, JavaScript, LaTeX
Spoken Languages: English (C1, TOEFL), French (A2, DELF)

AWARDS & HONORS

- 2022: Visiting Scholar Grant, Universidad del Rosario
- 2020: Research Seedbed Grant, Universidad del Rosario
- 2018: Visiting Scholar Grant, Universidad del Rosario
- 2018: Seed Capital Grant, Universidad del Rosario
- 2004: Huygens Scholarship for Outstanding Foreign Students, Netherlands Organization for International Cooperation in Higher Education